

Project Design Phase

Design Document

Field	Details
Project Name	[Your project name]
Prepared By	[Your name / team]
Version	1.0
Date	30 September 2026
Status	Draft

1. Introduction

The Design Phase turns the requirements gathered earlier into a clear blueprint for building the project. It defines what the system will look like, how its parts fit together, and how it will be built, so the development team can start coding with minimal confusion.

2. Objectives

- Convert requirements into a clear technical and functional design.
- Define system architecture, data flow and user interface.
- Identify risks early and plan how to handle them.
- Get stakeholder approval before development starts.

3. Scope

In scope: architecture design, database design, UI/UX wireframes, module breakdown, technology selection, test planning.

Out of scope: actual coding, deployment and post-release maintenance (covered in later phases).

4. System Architecture

The system follows a layered architecture so each part can be developed and tested independently:

- **Presentation layer:** web / mobile user interface.
- **Application layer:** business logic, APIs and services.
- **Data layer:** database and storage.
- **Integration layer:** third-party services and external APIs.

5. Module Design

Module	Description	Priority
User Management	Registration, login, roles and permissions	High
Core Feature	Main functionality of the project	High
Dashboard / Reports	Summary views and data reports	Medium
Notifications	Email / SMS / in-app alerts	Low

6. Database Design

Main entities are identified with their attributes and relationships (one-to-one, one-to-many). The design is normalised to reduce duplicate data, with primary and foreign keys defined for every table.

7. UI / UX Design

- Low-fidelity wireframes for every main screen.
- Consistent colour palette, fonts and components.
- Simple navigation with a maximum of three clicks to any feature.
- Responsive layout for mobile, tablet and desktop.

8. Technology Stack

Area	Choice
Frontend	[e.g. React / HTML, CSS, JS]
Backend	[e.g. Node.js / Python / Java]
Database	[e.g. MySQL / PostgreSQL / MongoDB]
Hosting	[e.g. AWS / Azure / Firebase]
Tools	Git, Figma, Jira / Trello

9. Security Design

- Authentication and role-based access control.
- Encryption of sensitive data in transit (HTTPS) and at rest.
- Input validation to prevent injection attacks.
- Regular backups and activity logging.

10. Risk Analysis

Risk	Impact	Mitigation
Requirement changes	High	Change control process and regular reviews
Schedule delays	Medium	Buffer time and weekly progress tracking
Technical complexity	Medium	Prototype early, choose familiar technologies
Budget overrun	High	Track costs per milestone

11. Timeline

Activity	Duration	Deliverable
Architecture design	1 week	Architecture diagram
Database design	1 week	ER diagram
UI/UX design	2 weeks	Wireframes and mockups
Review and approval	3 days	Signed-off design document

12. Deliverables

- System architecture document
- ER diagram and database schema
- UI wireframes / prototypes
- Test plan
- Approved design document

13. Approval

Role	Name	Signature	Date
Project Manager			
Client / Stakeholder			
Technical Lead			